Multi-linear Fit Results:

M Stars
Note: M Stars are limited to log M < 0.4 (M < 2.5)
RLM Relation:

$$\log R = \begin{cases} -0.10111 + 0.65022 \log L - 0.21105 \log M - 1.8808 (\log M)^2 & \log M \leq 0 \\ -0.23196 + 0.68899 \log L - 0.28028 \log M & \log M > 0 \end{cases}$$

Mean squared error: 0.00010649503066425212

Quadratic Part:
Intercept Std Error: 0.043
log L Std Error: 0.12
log M Std Error: 0.068
(log M)^2 Std Error: 0.404

```
                 OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.977
Model:                            OLS   Adj. R-squared:                  0.976
Method:                 Least Squares   F-statistic:                     925.6
Date:                Fri, 03 Jan 2020   Prob (F-statistic):           7.64e-54
Time:                        13:49:58   Log-Likelihood:                179.87
No. Observations:                  70   AIC:                            -351.7
Df Residuals:                      66   BIC:                            -342.7
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -0.1011      0.043     -2.362      0.021      -0.187      -0.016
x0             0.6502      0.012     52.313      0.000       0.625       0.675
x1            -0.2111      0.068     -3.115      0.003      -0.346      -0.076
x1Sq          -1.8808      0.404     -4.659      0.000      -2.687      -1.075
==============================================================================
Omnibus:                        4.186   Durbin-Watson:                   1.383
Prob(Omnibus):                  0.123   Jarque-Bera (JB):                4.889
Skew:                           0.071   Prob(JB):                       0.0868
Kurtosis:                       4.287   Cond. No.                         640.
==============================================================================

Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-0.10111306  0.65022046 -0.21105011 -1.88082381]

ANOVA results
              df    sum_sq   mean_sq            F        PR(>F)
x0           1.0  1.001240  1.001240  2750.191995  1.590797e-55
x1           1.0  0.001791  0.001791     4.920203  2.999041e-02
x1Sq         1.0  0.007902  0.007902    21.706120  1.588719e-05
Residual  66.0  0.024028  0.000364          NaN           NaN
```

Linear Part:
Intercept Std Error: 0.006
log L Std Error: 0.002
log M Std Error: 0.004

OLS Regression Results

Dep. Variable:	y	R-squared:	0.999
Model:	OLS	Adj. R-squared:	0.999
Method:	Least Squares	F-statistic:	1.072e+05
Date:	Fri, 03 Jan 2020	Prob (F-statistic):	0.00
Time:	13:51:29	Log-Likelihood:	1052.8
No. Observations:	306	AIC:	-2100.
Df Residuals:	303	BIC:	-2088.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-0.2407	0.006	-43.554	0.000	-0.252	-0.230
x0	0.6907	0.002	414.882	0.000	0.687	0.694
x1	-0.2649	0.004	-61.948	0.000	-0.273	-0.257

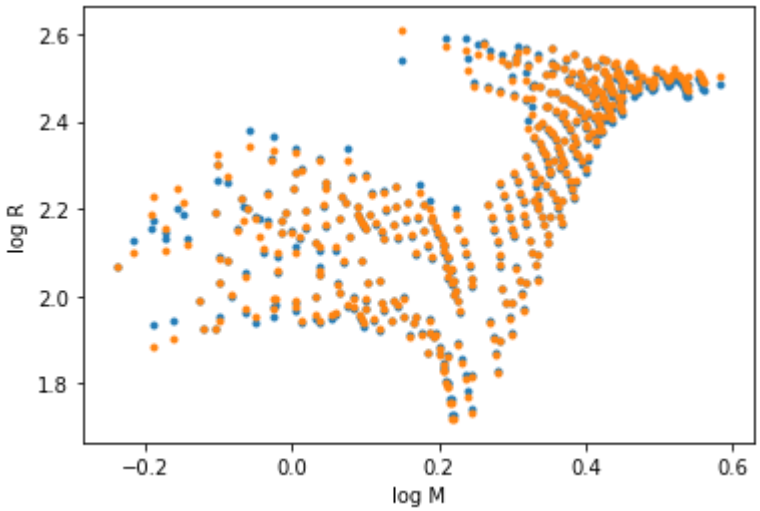
Omnibus:	133.073	Durbin-Watson:	1.692
Prob(Omnibus):	0.000	Jarque-Bera (JB):	7126.461
Skew:	-0.927	Prob(JB):	0.00
Kurtosis:	26.569	Cond. No.	52.7

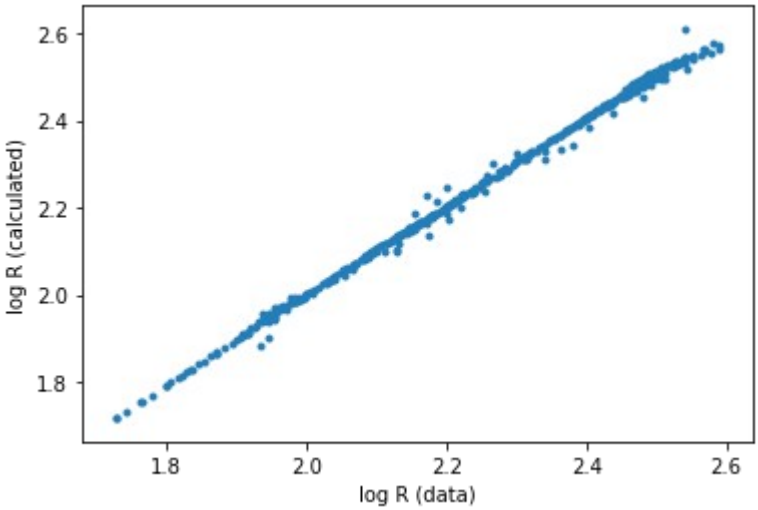
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

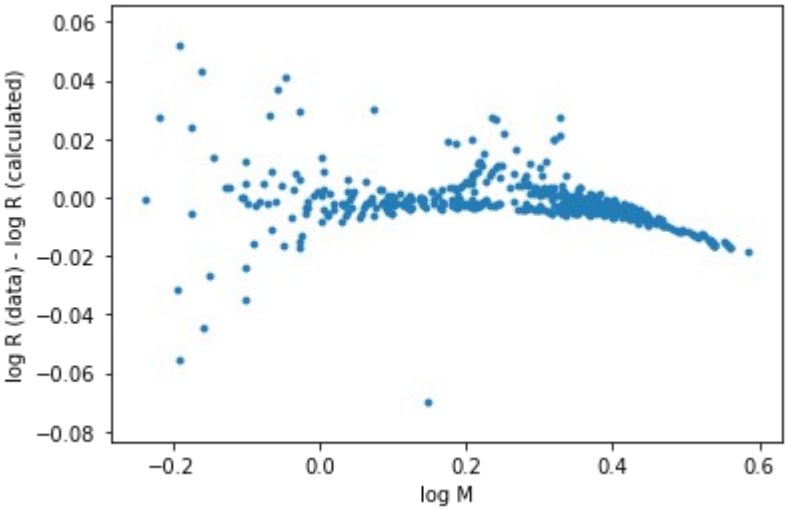
Retrieving manually the parameter estimates:
[-0.24069636 0.69073158 -0.26492684]

ANOVA results

	df	sum_sq	mean_sq	F	PR(>F)
x0	1.0	12.786718	12.786718	210591.088896	0.000000e+00
x1	1.0	0.233009	0.233009	3837.550371	4.280656e-174
Residual	303.0	0.018398	0.000061	NaN	NaN




log R (calculated)
log R (data)
2.6
2.4
2.2
2.0
1.8
1.8
2.0
2.2
2.4
2.6


log R (data) - log R (calculated)
log M
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.2
0.0
0.2
0.4
0.6

P0RM Relation:

$$\log P_0 = -2.05399 + 1.94312 \log R - 0.82834 \log M$$

Mean Squared Error: 0.0007128239264904945

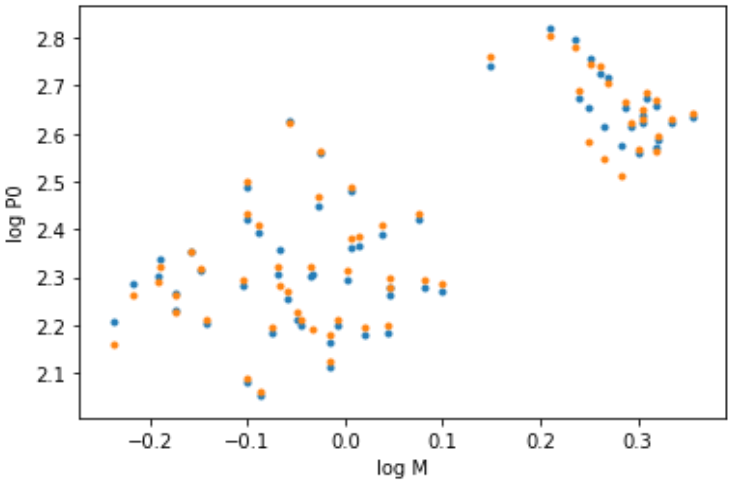
Intercept Std. Error: 0.121
log R Std. Error: 0.053
log M Std. Error: 0.051

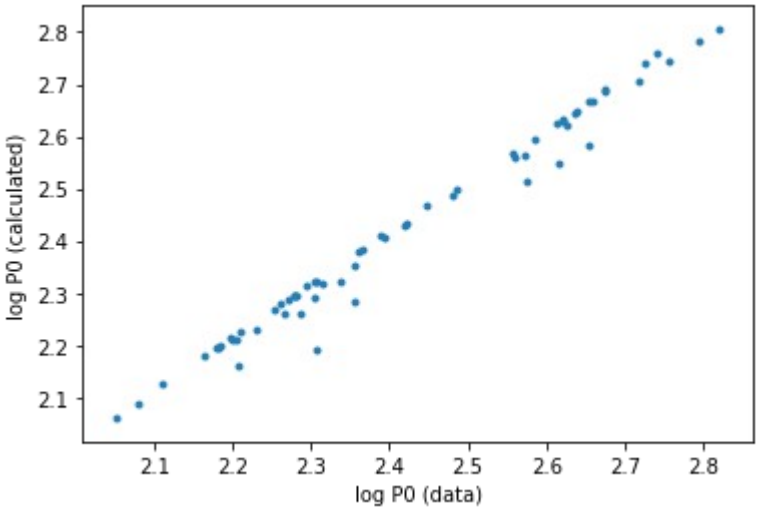
```
                 OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.983
Model:                            OLS   Adj. R-squared:                  0.982
Method:                 Least Squares   F-statistic:                     1664.
Date:                Fri, 03 Jan 2020   Prob (F-statistic):           1.29e-52
Time:                        13:57:27   Log-Likelihood:                136.66
No. Observations:                  62   AIC:                            -267.3
Df Residuals:                      59   BIC:                            -260.9
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -2.0540      0.121    -16.962      0.000      -2.296      -1.812
x0             1.9431      0.053     36.557      0.000       1.837       2.049
x1            -0.8283      0.051    -16.273      0.000      -0.930      -0.726
==============================================================================
Omnibus:                       46.140   Durbin-Watson:                   1.139
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              133.556
Skew:                           2.367   Prob(JB):                     9.97e-30
Kurtosis:                       8.412   Cond. No.                         103.
==============================================================================

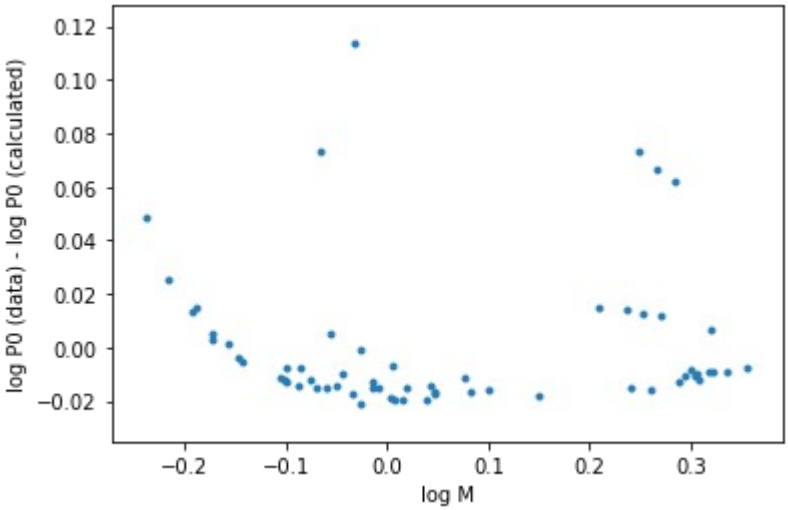
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-2.05399243  1.94312492 -0.82834175]

ANOVA results
              df    sum_sq   mean_sq            F        PR(>F)
x0           1.0  2.294472  2.294472  3063.096847  1.488148e-52
x1           1.0  0.198373  0.198373   264.825892  1.748719e-23
Residual    59.0  0.044195  0.000749          NaN           NaN
```


log P0
log M
2.8
2.7
2.6
2.5
2.4
2.3
2.2
2.1
−0.2
−0.1
0.0
0.1
0.2
0.3


log P0 (calculated)
log P0 (data)
2.8
2.7
2.6
2.5
2.4
2.3
2.2
2.1
2.1
2.2
2.3
2.4
2.5
2.6
2.7
2.8


log P0 (data) - log P0 (calculated)
log M
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.2
−0.1
0.0
0.1
0.2
0.3

P1RM Relation:

$$\log P_1 = -1.61247 + 1.59421 \log R - 0.52724 \log M$$

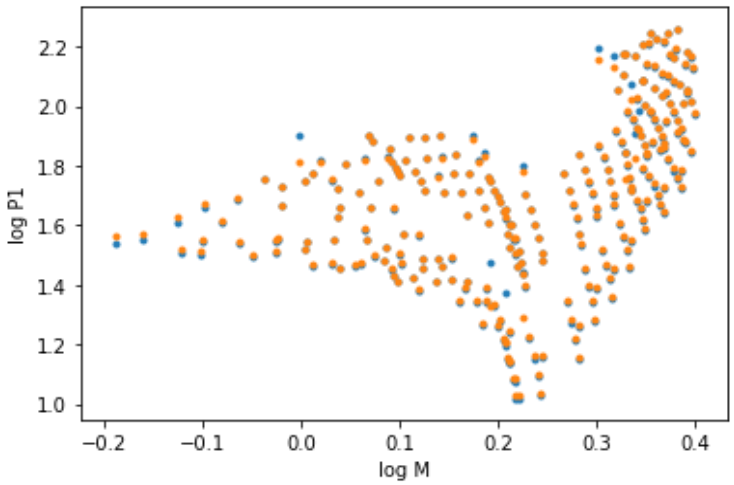
Mean Squared Error: 0.00041122633693616966
Intercept Std. Error: 0.016
logR Std. Error: 0.008
logM Std. Error: 0.011

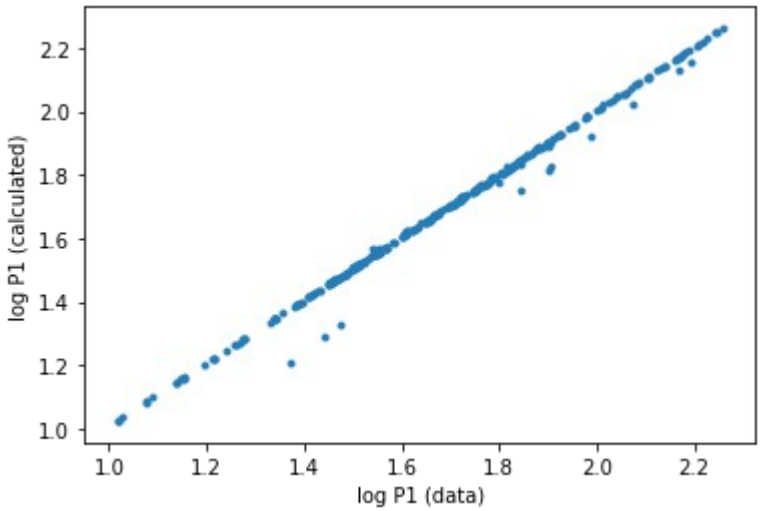
```
                 OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.995
Model:                            OLS   Adj. R-squared:                  0.994
Method:                 Least Squares   F-statistic:                 2.469e+04
Date:                Fri, 03 Jan 2020   Prob (F-statistic):          2.82e-308
Time:                        13:59:36   Log-Likelihood:                681.79
No. Observations:                 275   AIC:                            -1358.
Df Residuals:                     272   BIC:                            -1347.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -1.6226      0.016   -104.395      0.000      -1.653      -1.592
x0             1.5982      0.008    203.772      0.000       1.583       1.614
x1            -0.5161      0.011    -45.881      0.000      -0.538      -0.494
==============================================================================
Omnibus:                      355.191   Durbin-Watson:                   2.025
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            19913.112
Skew:                           6.048   Prob(JB):                         0.00
Kurtosis:                      42.894   Cond. No.                         36.0
==============================================================================

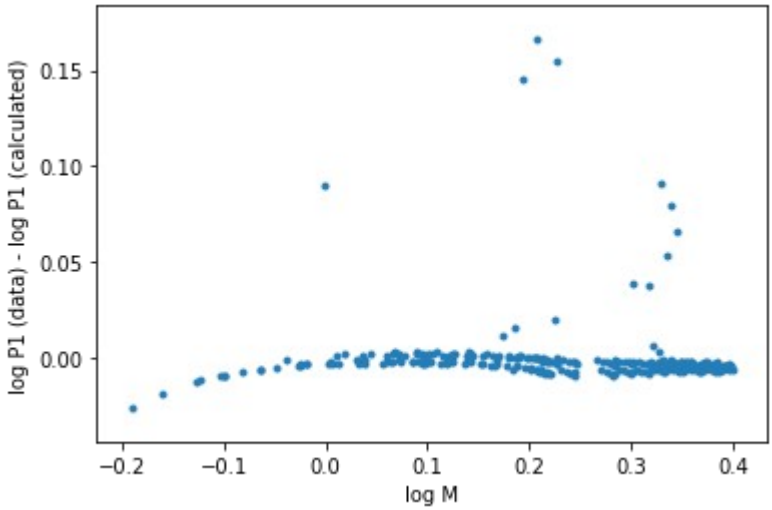
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-1.62255838  1.59820553 -0.51607087]

ANOVA results
                df     sum_sq    mean_sq             F         PR(>F)
x0          1.0  19.656568  19.656568  47278.423296  4.941597e-307
x1          1.0   0.875197   0.875197   2105.043384  4.685588e-130
Residual  272.0   0.113087   0.000416           NaN            NaN
```


log P1
log M
2.2
2.0
1.8
1.6
1.4
1.2
1.0
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4


log P1 (calculated)
log P1 (data)
2.2
2.0
1.8
1.6
1.4
1.2
1.0


log P1 (data) - log P1 (calculated)
log M
0.15
0.10
0.05
0.00
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4

C Stars

RLM Relation:

$$\log R = \begin{cases} -0.59530 + 0.81412 \log L + 0.22544 \log M - 3.1934 (\log M)^2, & \log M \leq 0.1 \\ -0.51330 + 0.80862 \log L - 0.71713 \log M, & \log M > 0.1 \end{cases}$$

Mean Squared Error: 0.00207013011688626

Quadratic Part:
Intercept Std. Error: 0.089
log L Std. Error: 0.024
log M Std. Error: 0.045
(log M)^2 Std. Error: 0.398

OLS Regression Results

Dep. Variable:	y	R-squared:	0.923
Model:	OLS	Adj. R-squared:	0.921
Method:	Least Squares	F-statistic:	457.9
Date:	Fri, 03 Jan 2020	Prob (F-statistic):	2.13e-63
Time:	14:16:16	Log-Likelihood:	194.87
No. Observations:	118	AIC:	-381.7
Df Residuals:	114	BIC:	-370.6
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-0.5953	0.089	-6.724	0.000	-0.771	-0.420
x0	0.8141	0.024	34.504	0.000	0.767	0.861
x1	0.2254	0.045	5.057	0.000	0.137	0.314
x1Sq	-3.1934	0.398	-8.026	0.000	-3.982	-2.405

Omnibus:	35.331	Durbin-Watson:	1.915
Prob(Omnibus):	0.000	Jarque-Bera (JB):	77.606
Skew:	1.195	Prob(JB):	1.41e-17
Kurtosis:	6.173	Cond. No.	350.

Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-0.59529744 0.81412451 0.22544442 -3.1933918]

ANOVA results

	df	sum_sq	mean_sq	F	PR(>F)
x0	1.0	2.911526	2.911526	1306.184907	2.820310e-64
x1	1.0	0.006693	0.006693	3.002534	8.583864e-02
x1Sq	1.0	0.143575	0.143575	64.411202	9.997384e-13
Residual	114.0	0.254109	0.002229	NaN	NaN

Linear Part:
Intercept Std. Error: 0.036
log L Std. Error: 0.010
log M Std. Error: 0.034

OLS Regression Results

Dep. Variable:	y	R-squared:	0.976
Model:	OLS	Adj. R-squared:	0.976
Method:	Least Squares	F-statistic:	3439.
Date:	Fri, 03 Jan 2020	Prob (F-statistic):	5.00e-137
Time:	14:17:22	Log-Likelihood:	380.76
No. Observations:	171	AIC:	-755.5
Df Residuals:	168	BIC:	-746.1
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-0.5133	0.036	-14.301	0.000	-0.584	-0.442
x0	0.8086	0.010	82.447	0.000	0.789	0.828
x1	-0.7171	0.034	-21.209	0.000	-0.784	-0.650

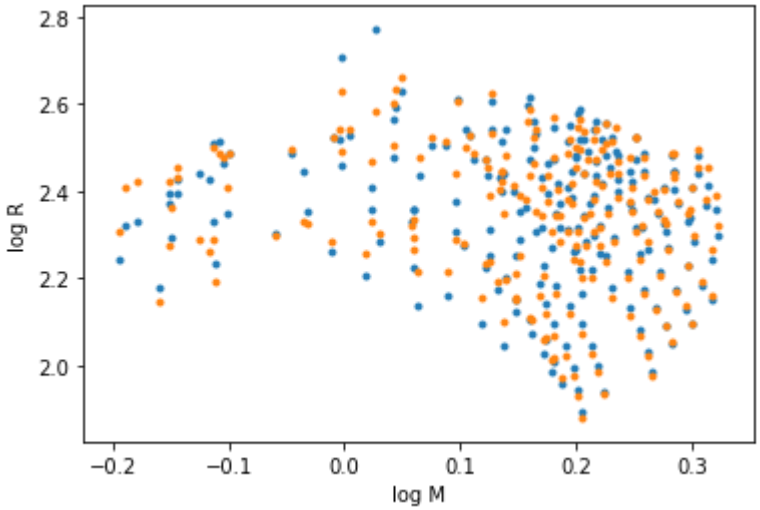
Omnibus:	12.147	Durbin-Watson:	2.298
Prob(Omnibus):	0.002	Jarque-Bera (JB):	18.742
Skew:	-0.391	Prob(JB):	8.51e-05
Kurtosis:	4.421	Cond. No.	70.9

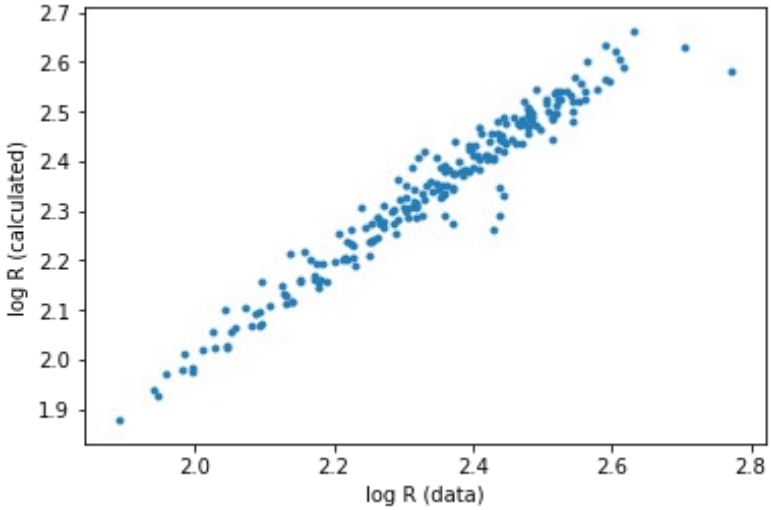
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

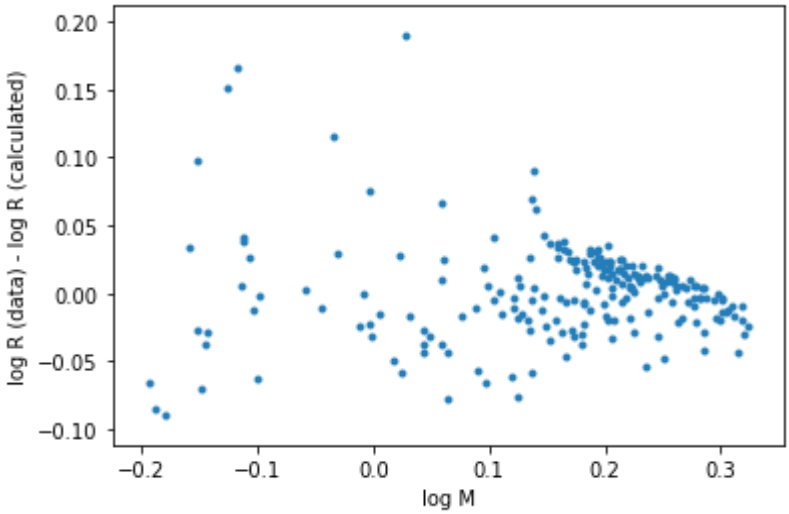
Retrieving manually the parameter estimates:
[-0.51329817 0.80861588 -0.71713426]

ANOVA results

	df	sum_sq	mean_sq	F	PR(>F)
x0	1.0	4.459221	4.459221	6428.140186	7.939711e-136
x1	1.0	0.312056	0.312056	449.840253	2.233459e-49
Residual	168.0	0.116542	0.000694	NaN	NaN


log R
log M
2.8
2.6
2.4
2.2
2.0
−0.2
−0.1
0.0
0.1
0.2
0.3


log R (calculated)
log R (data)
2.7
2.6
2.5
2.4
2.3
2.2
2.1
2.0
1.9
2.0
2.2
2.4
2.6
2.8


log R (data) - log R (calculated)
log M
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.2
−0.1
0.0
0.1
0.2
0.3

P0RM Relation:

$$\log P_0 = -1.10657 + 1.55879 \log R - 0.61466 \log M$$

Mean Squared Error: 0.004658883625203921

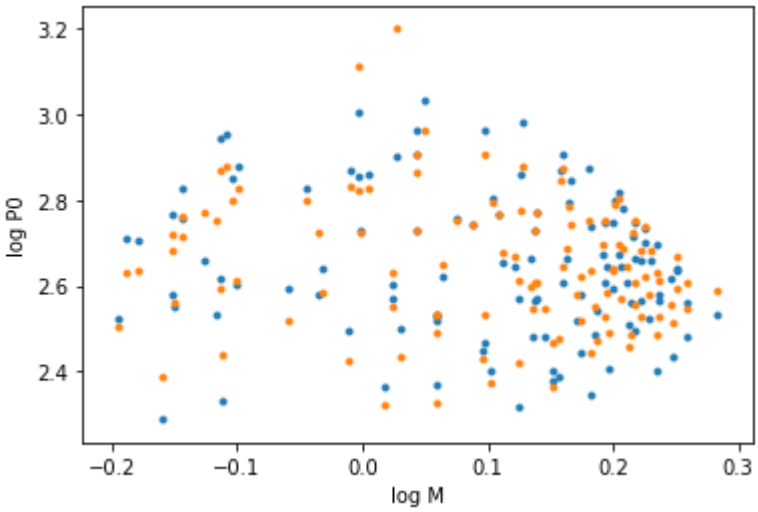
Intercept Std. Error: 0.160
log R Std. Error: 0.066
log M Std. Error: 0.054

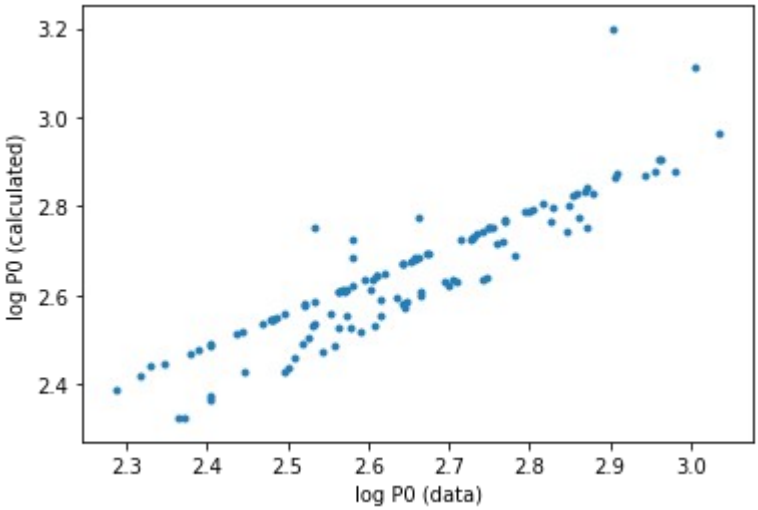
```
                 OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.835
Model:                            OLS   Adj. R-squared:                  0.832
Method:                 Least Squares   F-statistic:                     285.4
Date:                Fri, 03 Jan 2020   Prob (F-statistic):           6.74e-45
Time:                        14:23:22   Log-Likelihood:                146.80
No. Observations:                 116   AIC:                            -287.6
Df Residuals:                     113   BIC:                            -279.3
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -1.1066      0.160     -6.901      0.000      -1.424      -0.789
x0             1.5588      0.066     23.564      0.000       1.428       1.690
x1            -0.6147      0.054    -11.472      0.000      -0.721      -0.509
==============================================================================
Omnibus:                       25.916   Durbin-Watson:                   1.937
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               42.264
Skew:                          -1.017   Prob(JB):                     6.65e-10
Kurtosis:                       5.146   Cond. No.                         71.8
==============================================================================

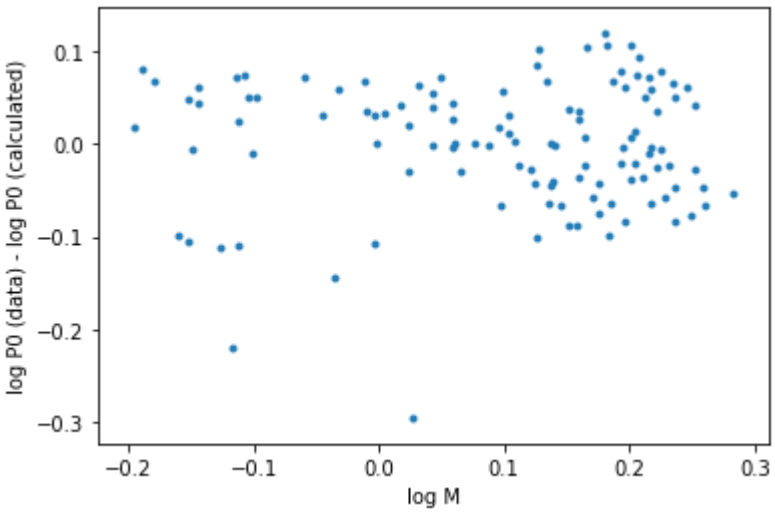
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-1.10657139  1.55879039 -0.61466354]

ANOVA results
                df    sum_sq   mean_sq           F        PR(>F)
x0             1.0  2.100055  2.100055  439.105817  9.951094e-41
x1             1.0  0.629440  0.629440  131.611251  1.137262e-20
Residual     113.0  0.540431  0.004783         NaN           NaN
```


log P0
log M
3.2
3.0
2.8
2.6
2.4
−0.2
−0.1
0.0
0.1
0.2
0.3


log P0 (calculated)
log P0 (data)
3.2
3.0
2.8
2.6
2.4
2.3
2.4
2.5
2.6
2.7
2.8
2.9
3.0


log P0 (data) - log P0 (calculated)
log M
0.1
0.0
−0.1
−0.2
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

P1RM Relation:

$$\log P_1 = -0.84669 + 1.29942 \log R - 0.56533 \log M$$

Mean Squared Error: 0.0017021860297208597

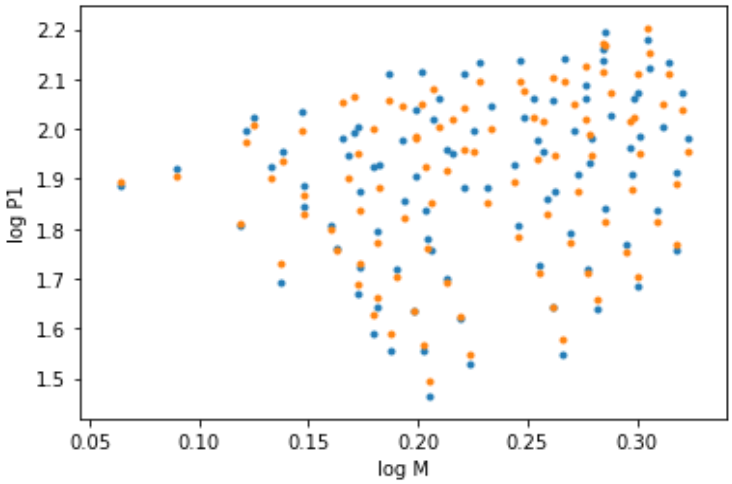
Intercept Std. Error: 0.071
log R Std. Error: 0.035
log M Std. Error: 0.079

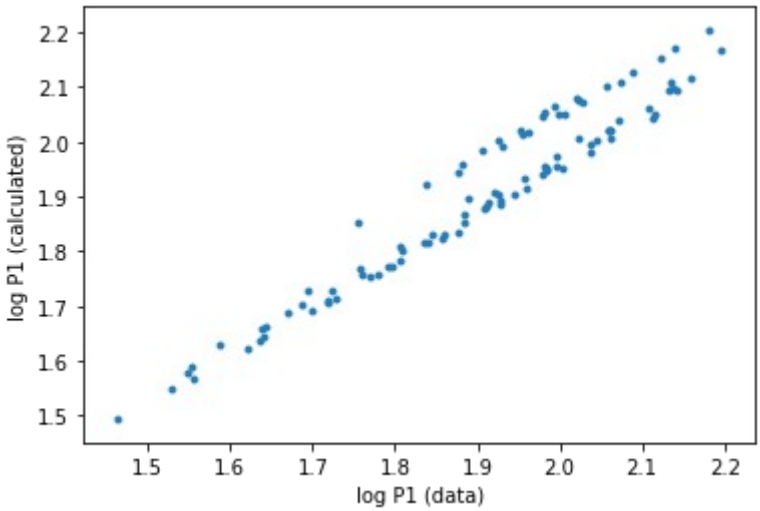
```
                 OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.941
Model:                            OLS   Adj. R-squared:                  0.940
Method:                 Least Squares   F-statistic:                     741.4
Date:                Fri, 03 Jan 2020   Prob (F-statistic):           7.10e-58
Time:                        14:26:07   Log-Likelihood:                 169.82
No. Observations:                  96   AIC:                            -333.6
Df Residuals:                      93   BIC:                            -326.0
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -0.8467      0.071    -11.845      0.000      -0.989      -0.705
x0             1.2994      0.035     37.507      0.000       1.231       1.368
x1            -0.5653      0.079     -7.126      0.000      -0.723      -0.408
==============================================================================
Omnibus:                       12.170   Durbin-Watson:                   2.395
Prob(Omnibus):                  0.002   Jarque-Bera (JB):                7.474
Skew:                          -0.527   Prob(JB):                       0.0238
Kurtosis:                       2.128   Cond. No.                         50.1
==============================================================================

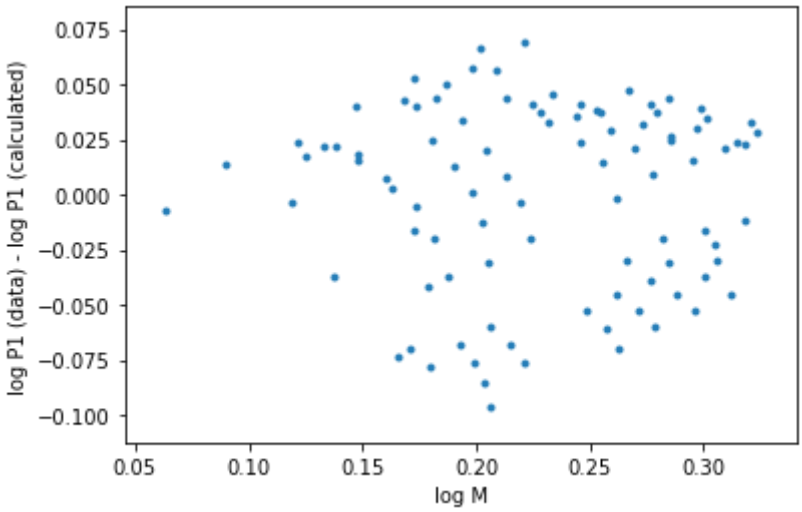
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Retrieving manually the parameter estimates:
[-0.84669133  1.29942476 -0.56533655]

ANOVA results
               df    sum_sq   mean_sq            F        PR(>F)
x0           1.0  2.516142  2.516142  1431.989664  2.769775e-58
x1           1.0  0.089224  0.089224    50.779076  2.168370e-10
Residual    93.0  0.163410  0.001757          NaN           NaN
```


log P1
log M
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
0.05
0.10
0.15
0.20
0.25
0.30


log P1 (calculated)
log P1 (data)
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5


log P1 (data) - log P1 (calculated)
log M
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
0.05
0.10
0.15
0.20
0.25
0.30